MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 7**

Inverse trigonometric and hyperbolic functions (Sections 6.6–6.7, Lecture 14); maximum and minimum values (Section 3.1, Lecture 15); the Mean Value Theorem and the shape of a graph (Sections 3.2–3.3, Lecture 16).

Key facts to have at hand (Lectures 14–16)

- **Inverse trigonometric functions — restrict first, then invert.** $\sin^{-1} x = y \iff \sin y = x$ and $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$ (domain $[-1,1]$); $\cos^{-1} x = y \iff \cos y = x$ and $0 \le y \le \pi$ (domain $[-1,1]$); $\tan^{-1} x = y \iff \tan y = x$ and $-\frac{\pi}{2} < y < \frac{\pi}{2}$ (domain $\mathbb{R}$). They are *not* $1/\sin x$, $1/\cos x$, $1/\tan x$. *Undoing:* $\sin^{-1}(\sin x) = x$ only for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$; $\cos^{-1}(\cos x) = x$ only on $[0,\pi]$; $\tan^{-1}(\tan x) = x$ only on $(-\frac{\pi}{2}, \frac{\pi}{2})$. $\lim_{x\to\pm\infty} \tan^{-1} x = \pm\frac{\pi}{2}$.
- **Their derivatives:** $\left(\sin^{-1} x\right)' = \dfrac{1}{\sqrt{1-x^2}}$, $\left(\cos^{-1} x\right)' = -\dfrac{1}{\sqrt{1-x^2}}$ $(-1 < x < 1$; vertical tangents at $\pm 1)$, $\left(\tan^{-1} x\right)' = \dfrac{1}{1+x^2}$ (every x); with the chain rule, u' on top and u inside.
- **Hyperbolic functions:** $\sinh x = \dfrac{e^x - e^{-x}}{2}$, $\cosh x = \dfrac{e^x + e^{-x}}{2}$, $\tanh x = \dfrac{\sinh x}{\cosh x}$; $\operatorname{sech} x = \dfrac{1}{\cosh x}$, $\operatorname{csch} x = \dfrac{1}{\sinh x}$, $\coth x = \dfrac{\cosh x}{\sinh x}$. $\cosh x \ge 1$; sinh is odd, cosh even; $-1 < \tanh x < 1$. *Identity:* $\cosh^2 x - \sinh^2 x = 1$. *Derivatives:* $(\sinh x)' = \cosh x$, $(\cosh x)' = \sinh x$ — *no minus sign*; $(\tanh x)' = \operatorname{sech}^2 x = 1 - \tanh^2 x$.
- **Extreme values.** $f(c)$ is the *absolute maximum* on D if $f(c) \ge f(x)$ for all x in D; a *local maximum* if $f(c) \ge f(x)$ for x near c (on an open interval around c — so never at an endpoint); minima likewise. *Extreme Value Theorem:* continuous on a closed interval $[a,b] \Rightarrow$ an absolute maximum and minimum exist. *Fermat:* a local maximum or minimum at c, and $f'(c)$ exists $\Rightarrow f'(c) = 0$; the converse is false (x^3), and an extreme value may sit where f' does not exist ($|x|$). *Critical point:* c in the domain with $f'(c) = 0$ or $f'(c)$ not existing — candidates, not answers.
- **The Closed Interval Method** (f continuous on $[a,b]$): the values of f at the critical points in (a,b) and at a and b; the largest is the absolute maximum, the smallest the absolute minimum.
- **Rolle** (f continuous on $[a,b]$, differentiable on (a,b), $f(a) = f(b)$): $f'(c) = 0$ for some c in (a,b). **The Mean Value Theorem** (f continuous on $[a,b]$, differentiable on (a,b)): $f(b) - f(a) = f'(c)(b-a)$ for some c in (a,b) — a tangent parallel to the secant. *Consequences:* $f' = 0$ on an interval $\Rightarrow f$ constant there; $f' = g'$ on an interval $\Rightarrow f = g + C$.
- **The shape of a graph.** *I/D Test:* $f' > 0$ on an interval $\Rightarrow f$ increasing there; $f' < 0 \Rightarrow$ decreasing. *First Derivative Test* (c a critical point of a continuous f): f' changes $+ \to -$ at c: local maximum; $- \to +$: local minimum; no change: neither. *Concavity:* the graph above its tangents = concave upward; below = concave downward. *Concavity Test:* $f'' > 0 \Rightarrow$ CU, $f'' < 0 \Rightarrow$ CD. An *inflection point* is where f is continuous and the concavity changes — found where f'' changes sign. *Second Derivative Test* (f'' continuous near c, $f'(c) = 0$): $f''(c) > 0 \Rightarrow$ local minimum; $f''(c) < 0 \Rightarrow$ local maximum; $f''(c) = 0 \Rightarrow$ *no information.*

Part I · Inverse trigonometric and hyperbolic functions; extreme values (6.6–6.7, 3.1) · 50 minutes

Warm-up I · True or false? (justify with one sentence or one counterexample)

1. $\sin^{-1}\left(\sin\dfrac{5\pi}{6}\right) = \dfrac{5\pi}{6}$.
2. $\cos^{-1}(\cos x) = x$ for every x in $[0,\pi]$.
3. $\tan^{-1} x = \dfrac{1}{\tan x}$.
4. $\sin^{-1} x$ is defined for every real number x.
5. $\lim\limits_{x\to-\infty} \tan^{-1} x = -\dfrac{\pi}{2}$.
6. $\dfrac{d}{dx}\cos^{-1} x = \dfrac{1}{\sqrt{1-x^2}}$ for $-1 < x < 1$.
7. $\cosh x \ge 1$ for every real x.

8. $\dfrac{d}{dx}\cosh x = -\sinh x$.
9. If f is continuous on the closed interval $[a,b]$, then f has an absolute maximum value on $[a,b]$.
10. If f has a local maximum at c, then $f'(c) = 0$.

Solution.

1 False. $\sin\dfrac{5\pi}{6} = \dfrac{1}{2}$, and $\sin^{-1}\dfrac{1}{2}$ is *the* number in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ whose sine is $\frac{1}{2}$: that is $\dfrac{\pi}{6}$, not $\dfrac{5\pi}{6}$. The undoing equation $\sin^{-1}(\sin x) = x$ holds only for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$.

2 True. cos was restricted to $[0,\pi]$ before it was inverted, so $\cos^{-1}(\cos x) = x$ exactly for $0 \le x \le \pi$.

3 False. The -1 in $\tan^{-1}$ means the *inverse function*, not the reciprocal. At $x = 1$: $\tan^{-1} 1 = \dfrac{\pi}{4} \approx 0.785$, whereas $\dfrac{1}{\tan 1} \approx 0.642$.

4 False. $\sin^{-1} x$ is the number whose sine is x, and $-1 \le \sin y \le 1$ for every y: the domain of $\sin^{-1}$ is $[-1,1]$. ($\sin^{-1} 2$ does not exist.)

5 True. $\lim\limits_{x\to-\infty} \tan^{-1} x = -\dfrac{\pi}{2}$: the line $y = -\frac{\pi}{2}$ is the horizontal asymptote of the graph of $\tan^{-1}$ on the left (and $y = \frac{\pi}{2}$ on the right).

6 False. $\dfrac{d}{dx}\cos^{-1} x = -\dfrac{1}{\sqrt{1-x^2}}$, with a minus sign — as the picture agrees: $\cos^{-1}$ falls from π to 0 across $[-1,1]$. (It is $\sin^{-1}$ whose derivative is $+\dfrac{1}{\sqrt{1-x^2}}$.)

7 True. $\cosh x - 1 = \dfrac{e^x + e^{-x}}{2} - 1 = \dfrac{e^x - 2 + e^{-x}}{2} = \dfrac{1}{2}\left(e^{x/2} - e^{-x/2}\right)^2 \ge 0$, with equality only at $x = 0$. The range of cosh is $[1,\infty)$.

8 False. $\dfrac{d}{dx}\cosh x = \dfrac{e^x - e^{-x}}{2} = \sinh x$ — with no minus sign. The minus sign belongs to $(\cos x)' = -\sin x$.

9 True. This is the Extreme Value Theorem: continuous on a *closed* interval guarantees an absolute maximum (and an absolute minimum).

10 False. Fermat's Theorem says: a local maximum or minimum at c *and* $f'(c)$ *exists* force $f'(c) = 0$. Without the second condition it fails: $f(x) = -|x|$ has a local (even absolute) maximum at 0, since $-|x| \le 0 = f(0)$, but $f'(0)$ does not exist — the slopes are 1 from the left and -1 from the right.

Problem 1 *Inverse trigonometric functions: values, the quadrant, simplification.*

(a) Find the exact values: $\sin^{-1}\left(-\dfrac{\sqrt{3}}{2}\right)$, $\cos^{-1}\left(-\dfrac{\sqrt{2}}{2}\right)$, $\tan^{-1}\left(-\sqrt{3}\right)$, $\sin^{-1}\left(\sin\dfrac{3\pi}{4}\right)$.

(b) Find the exact values of $\tan\left(\cos^{-1}\dfrac{7}{25}\right)$ and $\cos\left(\tan^{-1}(-2)\right)$. Draw a right triangle for each — and decide the sign from the *range* of the inverse function before you write the answer.

(c) Simplify $\sin\left(2\tan^{-1} x\right)$ for every real x, and $\cos\left(2\sin^{-1} x\right)$ for $-1 \le x \le 1$.

Solution.

(a) Each answer must lie in the range of the inverse function — that is what makes it *the* answer. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\dfrac{\pi}{3}$: $\sin\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$ and $-\frac{\pi}{3}$ is in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$. (The number $\frac{4\pi}{3}$ also has sine $-\frac{\sqrt{3}}{2}$, but it is not in the range.) $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right) = \dfrac{3\pi}{4}$: $\cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$ and $\frac{3\pi}{4}$ is in $[0,\pi]$. $\tan^{-1}\left(-\sqrt{3}\right) = -\dfrac{\pi}{3}$: $\tan\left(-\frac{\pi}{3}\right) = -\sqrt{3}$ and $-\frac{\pi}{3}$ is in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$. $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) = \dfrac{\pi}{4}$: $\sin\frac{3\pi}{4} = \frac{\sqrt{2}}{2}$, and the number in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ with sine $\frac{\sqrt{2}}{2}$ is $\frac{\pi}{4}$. The answer is *not* $\frac{3\pi}{4}$: undoing works only on $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.

(b) Let $\theta = \cos^{-1}\frac{7}{25}$, so $\cos\theta = \frac{7}{25}$ and $0 \le \theta \le \pi$; since the cosine is positive, $0 < \theta < \frac{\pi}{2}$. A right triangle with adjacent side 7 and hypotenuse 25 has opposite side $\sqrt{25^2 - 7^2} = \sqrt{576} = 24$,

so $\tan\theta = \dfrac{24}{7}$ — positive, as it must be in the first quadrant. Let $\varphi = \tan^{-1}(-2)$, so $\tan\varphi = -2$ and $-\frac{\pi}{2} < \varphi < \frac{\pi}{2}$; since the tangent is negative, $-\frac{\pi}{2} < \varphi < 0$, where the *cosine is positive*. From $1+\tan^2\varphi = \sec^2\varphi$: $\sec^2\varphi = 5$, so $\cos^2\varphi = \frac{1}{5}$ and, taking the positive root, $\cos\varphi = \dfrac{1}{\sqrt{5}}$. (The triangle has legs 1 and 2 and hypotenuse $\sqrt{5}$; the triangle gives $\frac{1}{\sqrt{5}}$ up to sign, and the range of $\tan^{-1}$ decides the sign.)

(c) Let $\theta = \tan^{-1}x$, so $\tan\theta = x$ and $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. By the double-angle formula, $\sin 2\theta = 2\sin\theta\cos\theta$; write $\sin\theta = \tan\theta\cos\theta$ and use $\cos^2\theta = \dfrac{1}{\sec^2\theta} = \dfrac{1}{1+\tan^2\theta}$:

$$\sin\bigl(2\tan^{-1}x\bigr) = 2\tan\theta\cos^2\theta = \frac{2\tan\theta}{1+\tan^2\theta} = \frac{2x}{1+x^2}.$$

No sign question arose, because only $\cos^2\theta$ was needed. (Check at $x=1$: $\sin\bigl(2\cdot\frac{\pi}{4}\bigr) = 1$ and $\frac{2}{1+1} = 1$.) For the second, let $\theta = \sin^{-1}x$, so $\sin\theta = x$: $\cos\bigl(2\sin^{-1}x\bigr) = \cos 2\theta = 1 - 2\sin^2\theta = 1 - 2x^2$ — again no square root.

Problem 2 *Derivatives with inverse trigonometric functions.*

Differentiate. For every $\sin^{-1}(\dots)$, $\cos^{-1}(\dots)$ or $\tan^{-1}(\dots)$ say what the *inside* is before you write anything.

(a) $y = \sin^{-1}(5x)$

(b) $y = x^2\tan^{-1}(4x)$

(c) $y = \cos^{-1}\bigl(e^{-x}\bigr)$, $x > 0$

(d) $y = \tan^{-1}\dfrac{x}{3}$

(e) $y = \sin^{-1}x + x\sqrt{1-x^2}$

(f) The tangent line to $y = \cos^{-1}\dfrac{x}{3}$ at $x = \dfrac{3}{2}$.

Solution. The rules, with the chain rule — the outside's derivative evaluated at the inside, times the inside's derivative:

$$\bigl(\sin^{-1}u\bigr)' = \frac{u'}{\sqrt{1-u^2}}, \qquad \bigl(\cos^{-1}u\bigr)' = -\frac{u'}{\sqrt{1-u^2}}, \qquad \bigl(\tan^{-1}u\bigr)' = \frac{u'}{1+u^2}.$$

(a) Inside $u = 5x$, $u' = 5$: $y' = \dfrac{5}{\sqrt{1-(5x)^2}} = \dfrac{5}{\sqrt{1-25x^2}}$, for $|5x| < 1$, i.e. $-\frac{1}{5} < x < \frac{1}{5}$.

(b) A product: $y' = (x^2)'\tan^{-1}(4x) + x^2\bigl(\tan^{-1}(4x)\bigr)'$. Inside $u = 4x$, $u' = 4$: $\bigl(\tan^{-1}(4x)\bigr)' = \dfrac{4}{1+(4x)^2} = \dfrac{4}{1+16x^2}$. So

$$y' = 2x\tan^{-1}(4x) + \frac{4x^2}{1+16x^2}.$$

(The two usual slips: forgetting the inside's 4, and writing $\frac{1}{1+x^2}$ — the outside's derivative at x instead of at the inside $4x$.)

(c) Inside $u = e^{-x}$, $u' = -e^{-x}$:

$$y' = -\frac{u'}{\sqrt{1-u^2}} = -\frac{-e^{-x}}{\sqrt{1-e^{-2x}}} = \frac{e^{-x}}{\sqrt{1-e^{-2x}}}.$$

This needs $|e^{-x}| < 1$, i.e. $x > 0$; at $x = 0$ the inside is 1, $y = \cos^{-1}1 = 0$, and the graph has a vertical tangent. The derivative is positive: y increases from 0 (at $x = 0$) towards $\cos^{-1}0 = \frac{\pi}{2}$ as $x \to \infty$.

(d) Inside $u = \dfrac{x}{3}$, $u' = \dfrac{1}{3}$: $y' = \dfrac{1/3}{1+x^2/9} = \dfrac{1/3}{(9+x^2)/9} = \dfrac{3}{9+x^2}$.

(e) Three pieces: $\big(\sin^{-1}x\big)' = \dfrac{1}{\sqrt{1-x^2}}$, and the product $x\sqrt{1-x^2}$ has derivative $\sqrt{1-x^2} + x\cdot\dfrac{-2x}{2\sqrt{1-x^2}} = \sqrt{1-x^2} - \dfrac{x^2}{\sqrt{1-x^2}}$. Put everything over $\sqrt{1-x^2}$:

$$y' = \frac{1+(1-x^2)-x^2}{\sqrt{1-x^2}} = \frac{2-2x^2}{\sqrt{1-x^2}} = \frac{2(1-x^2)}{\sqrt{1-x^2}} = 2\sqrt{1-x^2} \qquad (-1<x<1).$$

(Check at $x=0$: $y'(0) = 1+1-0 = 2$ from the three pieces, and $2\sqrt{1-0} = 2$.)

(f) The point: $y\big(\frac{3}{2}\big) = \cos^{-1}\dfrac{3/2}{3} = \cos^{-1}\dfrac{1}{2} = \dfrac{\pi}{3}$. The slope: inside $u = \dfrac{x}{3}$, $u' = \dfrac{1}{3}$, so $y' = -\dfrac{1/3}{\sqrt{1-x^2/9}} = -\dfrac{1/3}{\sqrt{(9-x^2)/9}} = -\dfrac{1/3}{\sqrt{9-x^2}/3} = -\dfrac{1}{\sqrt{9-x^2}}$, and $y'\big(\frac{3}{2}\big) = -\dfrac{1}{\sqrt{9-9/4}} = -\dfrac{1}{\sqrt{27/4}} = -\dfrac{2}{\sqrt{27}} = -\dfrac{2}{3\sqrt{3}}$. The tangent line: $y - \dfrac{\pi}{3} = -\dfrac{2}{3\sqrt{3}}\Big(x-\dfrac{3}{2}\Big)$, that is $y = \dfrac{\pi}{3} - \dfrac{2}{3\sqrt{3}}\Big(x-\dfrac{3}{2}\Big)$ $(\approx 1.047 - 0.385(x-1.5))$. A negative slope, as for every tangent to $\cos^{-1}$.

Problem 3 *Hyperbolic functions: values and identities (homework).*

(a) Find the exact values of $\cosh\big(\ln\frac{5}{2}\big)$, $\sinh\big(\ln\frac{5}{2}\big)$ and $\tanh\big(\ln\frac{5}{2}\big)$; then check that $\cosh^2\big(\ln\frac{5}{2}\big) - \sinh^2\big(\ln\frac{5}{2}\big) = 1$.

(b) Prove, from the definitions, that $\cosh 2x = 1 + 2\sinh^2 x$ for every x.

(c) Show that $\cosh(\ln x) = \dfrac{x^2+1}{2x}$ for $x > 0$.

(d) Solve the equation $\cosh x = 2$. (Write it in terms of e^x.) How many solutions are there, and why was that to be expected?

Solution.

(a) Since $e^{\ln(5/2)} = \dfrac{5}{2}$ and $e^{-\ln(5/2)} = \dfrac{2}{5}$ (undoing), the definitions give

$$\cosh\Big(\ln\frac{5}{2}\Big) = \frac{\frac{5}{2}+\frac{2}{5}}{2} = \frac{29/10}{2} = \frac{29}{20}, \qquad \sinh\Big(\ln\frac{5}{2}\Big) = \frac{\frac{5}{2}-\frac{2}{5}}{2} = \frac{21/10}{2} = \frac{21}{20},$$

and then $\tanh\big(\ln\frac{5}{2}\big) = \dfrac{\sinh(\ln\frac{5}{2})}{\cosh(\ln\frac{5}{2})} = \dfrac{21/20}{29/20} = \dfrac{21}{29}$. Check: $\cosh^2\big(\ln\frac{5}{2}\big) - \sinh^2\big(\ln\frac{5}{2}\big) = \dfrac{841}{400} - \dfrac{441}{400} = \dfrac{400}{400} = 1$. (The point $(\cosh t, \sinh t) = \big(\frac{29}{20}, \frac{21}{20}\big)$ lies on the hyperbola $x^2 - y^2 = 1$; and $20^2 + 21^2 = 29^2$.)

(b) Proof. From the definition, $\sinh^2 x = \Big(\dfrac{e^x - e^{-x}}{2}\Big)^2 = \dfrac{e^{2x} - 2 + e^{-2x}}{4}$. Hence

$$1 + 2\sinh^2 x = 1 + \frac{e^{2x}-2+e^{-2x}}{2} = 1 + \frac{e^{2x}+e^{-2x}}{2} - 1 = \frac{e^{2x}+e^{-2x}}{2} = \cosh 2x. \qquad \square$$

(Compare $\cos 2x = 1 - 2\sin^2 x$: the same shape, the sign changed — as in $\cosh^2 x - \sinh^2 x = 1$ against $\cos^2 x + \sin^2 x = 1$.)

(c) $e^{\ln x} = x$ and $e^{-\ln x} = \dfrac{1}{x}$ for $x > 0$, so $\cosh(\ln x) = \dfrac{x + \frac{1}{x}}{2} = \dfrac{\frac{x^2+1}{x}}{2} = \dfrac{x^2+1}{2x}$. $\square$ (At $x = \frac{5}{2}$: $\dfrac{25/4+1}{5} = \dfrac{29/4}{5} = \dfrac{29}{20}$, as in (a).)

(d) $\cosh x = 2$ means $\dfrac{e^x + e^{-x}}{2} = 2$, i.e. $e^x + e^{-x} = 4$. Multiply by $e^x > 0$: $(e^x)^2 - 4e^x + 1 = 0$, a quadratic in $u = e^x$, with roots $u = \dfrac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$. Both are positive ($\sqrt{3} < 2$), so both are allowed: $e^x = 2+\sqrt{3}$ or $e^x = 2-\sqrt{3}$, that is

$$x = \ln\big(2+\sqrt{3}\big) \approx 1.317 \qquad \text{or} \qquad x = \ln\big(2-\sqrt{3}\big) = -\ln\big(2+\sqrt{3}\big) \approx -1.317,$$

the second because $(2-\sqrt{3})(2+\sqrt{3}) = 1$, so $2-\sqrt{3} = \dfrac{1}{2+\sqrt{3}}$. Two solutions, negatives of each other: that was to be expected, because cosh is even ($\cosh(-x) = \cosh x$) — it takes every value > 1 twice, so it is not one-to-one, and "$\cosh^{-1} 2$" would need a restriction to $x \geq 0$ first.

Problem 4 *Derivatives of hyperbolic functions.*

(a) $y = \cosh\big(x^3\big)$

(b) $y = \ln(1+\cosh x)$

(c) $y = \tanh\big(x^2\big)$

(d) $y = \dfrac{\sinh x}{1+\cosh x}$ (simplify the answer)

(e) Prove two of the three formulas of the lecture's table that were left as homework there:

$$\frac{d}{dx}\operatorname{sech} x = -\operatorname{sech} x \tanh x \qquad \text{and} \qquad \frac{d}{dx}\coth x = -\operatorname{csch}^2 x.$$

Solution. The rules: $(\sinh u)' = \cosh u \cdot u'$, $(\cosh u)' = \sinh u \cdot u'$ (no minus sign), $(\tanh u)' = \operatorname{sech}^2 u \cdot u'$.

(a) Inside $u = x^3$, $u' = 3x^2$: $y' = \sinh(x^3) \cdot 3x^2 = 3x^2 \sinh(x^3)$ — no minus sign.

(b) $\big(\ln u\big)' = \dfrac{u'}{u}$ with $u = 1+\cosh x$, which is ≥ 2, so the logarithm is defined for every x (no "$x > 0$" needed, unlike $\ln(\sinh x)$); and $u' = \sinh x$: $y' = \dfrac{\sinh x}{1+\cosh x}$ — exactly the function of part (d), so (d) computes y'' for this y.

(c) Inside $u = x^2$, $u' = 2x$: $y' = \operatorname{sech}^2(x^2) \cdot 2x = 2x \operatorname{sech}^2(x^2)$ $\big(= 2x\big(1 - \tanh^2(x^2)\big)\big)$.

(d) Quotient rule, with $(\sinh x)' = \cosh x$ and $(1+\cosh x)' = \sinh x$:

$$y' = \frac{\cosh x\,(1+\cosh x) - \sinh x \cdot \sinh x}{(1+\cosh x)^2} = \frac{\cosh x + \cosh^2 x - \sinh^2 x}{(1+\cosh x)^2}.$$

Now the identity $\cosh^2 x - \sinh^2 x = 1$ turns the numerator into $\cosh x + 1$, which cancels one factor of the denominator:

$$y' = \frac{1+\cosh x}{(1+\cosh x)^2} = \frac{1}{1+\cosh x}.$$

(Without the identity the answer would stay a mess. A remark: $\dfrac{\sinh x}{1+\cosh x} = \tanh\dfrac{x}{2}$, so this is also $\frac{1}{2}\operatorname{sech}^2\frac{x}{2}$ — the same thing.)

(e) Proof. $\operatorname{sech} x = \dfrac{1}{\cosh x} = (\cosh x)^{-1}$; by the chain rule, with inside $\cosh x$ and $(\cosh x)' = \sinh x$,

$$\frac{d}{dx}\operatorname{sech} x = -(\cosh x)^{-2} \cdot \sinh x = -\frac{1}{\cosh x} \cdot \frac{\sinh x}{\cosh x} = -\operatorname{sech} x \tanh x.$$

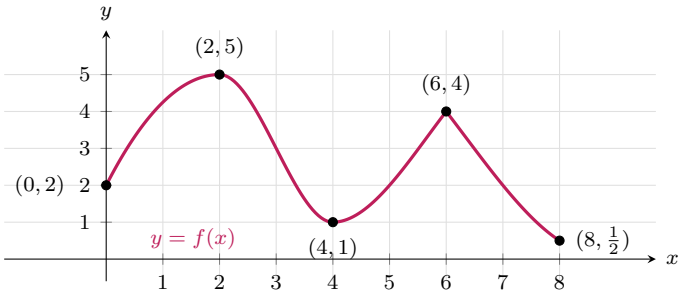
$\coth x = \dfrac{\cosh x}{\sinh x}$; by the quotient rule and then the identity,

$$\frac{d}{dx}\coth x = \frac{\sinh x \cdot \sinh x - \cosh x \cdot \cosh x}{\sinh^2 x} = \frac{-(\cosh^2 x - \sinh^2 x)}{\sinh^2 x} = -\frac{1}{\sinh^2 x} = -\operatorname{csch}^2 x$$

for $x \neq 0$. □ (Beside trigonometry: $(\sec x)' = \sec x \tan x$ and $(\cot x)' = -\csc^2 x$ — the hyperbolic sech gets a minus sign that sec does not have, because $(\cosh x)' = +\sinh x$.)

Problem 5 *Extreme values from a graph; the Extreme Value Theorem (homework).*

(a) The graph of a function f with domain $[0, 8]$ is shown. State the absolute maximum and minimum values and where they are taken; all the local maximum and minimum values; and the critical points of f in $(0, 8)$. At which of the five marked points does Fermat's Theorem apply, and what does it say there?



(b) For each function, say whether the Extreme Value Theorem *guarantees* an absolute maximum and an absolute minimum. If it does not, say which hypothesis fails, and whether the extreme values exist anyway.

(i) $f(x) = x^2 - 6x$ on $[0, 7]$;

(ii) $g(x) = \dfrac{1}{x}$ on $(0, 1]$;

(iii) $h(x) = x^2$ for $0 \le x < 2$ and $h(2) = 1$, on $[0, 2]$.

Solution.

(a) *Absolute:* the largest value on the whole domain is $f(2) = 5$, so the absolute maximum is 5, at $x = 2$; the smallest is $f(8) = \frac{1}{2}$, so the absolute minimum is $\frac{1}{2}$, at the endpoint $x = 8$. *Local:* "local" means largest or smallest on some open interval around the point, so an endpoint can never be local. $f(2) = 5$ is a local maximum (and the absolute one); $f(6) = 4$ is a local maximum — the graph has a corner there, but the definition does not care; $f(4) = 1$ is a local minimum. $f(0) = 2$ and $f(8) = \frac{1}{2}$ are not local extrema. *Critical points in* $(0, 8)$*:* where $f' = 0$ (horizontal tangents): $x = 2$ and $x = 4$; where f' does not exist: the corner $x = 6$. So 2, 4, 6. (The endpoints 0 and 8 are treated separately, as endpoints — that is what the Closed Interval Method does with them.) *Fermat's Theorem* needs a local maximum or minimum at c *and* the existence of $f'(c)$. That is the case at 2 and at 4: it says $f'(2) = 0$ and $f'(4) = 0$, which the horizontal tangents confirm. At 6 there is a local maximum, but $f'(6)$ does not exist, so the theorem says nothing (and indeed there is no horizontal tangent). At 0 and 8 there is no local extremum, so the theorem does not apply either — the absolute minimum at 8 comes with a non-zero one-sided slope.

(b)(i) f is a polynomial, continuous everywhere, and $[0, 7]$ is closed: the theorem guarantees both. Here $f(x) = (x - 3)^2 - 9$: minimum -9 at $x = 3$, maximum $f(7) = 7$ at the endpoint.

(b)(ii) g is continuous on $(0, 1]$, but the interval is not closed: no guarantee. In fact there is *no maximum*: $\frac{1}{x}$ grows without bound as $x \to 0^+$. A minimum exists anyway: $g(1) = 1$, since $\frac{1}{x} \ge 1$ on $(0, 1]$. (One hypothesis failing does not forbid an extreme value; it only withdraws the promise.)

(b)(iii) The interval $[0, 2]$ is closed, but h is not continuous at 2 (the values approach 4, and $h(2) = 1$): no guarantee. The values of h are $[0, 4) \cup \{1\}$: there is *no maximum* — the values come as close to 4 as we like but never reach it — while the minimum 0 is taken at $x = 0$.

Problem 6 *Critical points.*

Find all the critical points of each function — the numbers in the domain where f' is 0 *or does not exist.*

(a) $f(x) = 2x^3 - 9x^2 - 24x + 5$

(b) $g(x) = \big(x^2 - 4\big)^{2/3}$

(c) $h(x) = 3\tan^{-1} x - x$

(d) $k(x) = x\, e^{-x^2}$

Solution.

(a) $f'(x) = 6x^2 - 18x - 24 = 6(x^2 - 3x - 4) = 6(x+1)(x-4)$, which exists everywhere; so the critical points are the zeros of f': $x = -1$ and $x = 4$.

(b) $g(x) = \big(x^2 - 4\big)^{2/3} = \sqrt[3]{(x^2-4)^2}$ is defined for every x (a cube root of a square): the domain is $\mathbb{R}$. The chain rule, outside $u^{2/3}$ and inside $u = x^2 - 4$, gives for $x^2 - 4 \neq 0$

$$g'(x) = \frac{2}{3}\big(x^2-4\big)^{-1/3} \cdot 2x = \frac{4x}{3\sqrt[3]{x^2-4}},$$

which is 0 exactly at $x = 0$. At $x = \pm 2$ the formula does not apply (division by 0), so we go back to the definition. At 2, with $g(2) = 0$:

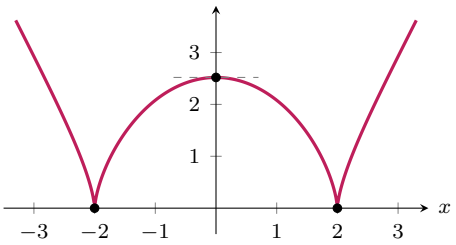
$$\lim_{h\to 0} \frac{g(2+h) - g(2)}{h} = \lim_{h\to 0} \frac{\big((2+h)^2 - 4\big)^{2/3}}{h} = \lim_{h\to 0} \frac{\big(h(4+h)\big)^{2/3}}{h}$$
$$= \lim_{h\to 0} \frac{h^{2/3}(4+h)^{2/3}}{h} = \lim_{h\to 0} \frac{(4+h)^{2/3}}{\sqrt[3]{h}},$$

and $\lim\limits_{h\to 0^+} \dfrac{(4+h)^{2/3}}{\sqrt[3]{h}} = +\infty$ while $\lim\limits_{h\to 0^-} \dfrac{(4+h)^{2/3}}{\sqrt[3]{h}} = -\infty$ (the numerator tends to $4^{2/3} > 0$, the denominator to 0 through positive, then negative, values): the limit does not exist, so $g'(2)$ does not exist: the graph has a cusp at $(2, 0)$, the slope running off to $-\infty$ from the left and to $+\infty$ from the right. Since $g(-x) = g(x)$, the same happens at -2. Both ± 2 are in the domain, so they are critical points. *Critical points:* -2, 0, 2. (Values: $g(\pm 2) = 0$, the smallest possible since $g \geq 0$; $g(0) = \sqrt[3]{16} = 2\sqrt[3]{2} \approx 2.52$.)

(c) $h'(x) = \dfrac{3}{1+x^2} - 1 = \dfrac{3 - (1+x^2)}{1+x^2} = \dfrac{2-x^2}{1+x^2}$, defined everywhere, zero exactly when $x^2 = 2$. Critical points: $-\sqrt{2}$ and $\sqrt{2}$. ($h' > 0$ between them and $h' < 0$ outside; $h(\pm\sqrt{2}) = \pm(3\tan^{-1}\sqrt{2} - \sqrt{2}) \approx \pm 1.452$.)

(d) Product rule, with $\big(e^{-x^2}\big)' = -2x\, e^{-x^2}$ (inside $-x^2$): $k'(x) = 1\cdot e^{-x^2} + x\cdot(-2x)e^{-x^2} = (1-2x^2)e^{-x^2}$. Since $e^{-x^2} > 0$, $k' = 0$ exactly when $2x^2 = 1$. Critical points: $-\dfrac{1}{\sqrt{2}}$ and $\dfrac{1}{\sqrt{2}}$.

6(b): $y = (x^2-4)^{2/3}$ — cusps at ± 2, a horizontal tangent at 0



Problem 7 *The Closed Interval Method.*

Find the absolute maximum and minimum values of each function on the given interval, and where they are taken.

(a) $f(x) = 2x^3 - 15x^2 + 24x + 3$ on $[0, 3]$

(b) $g(x) = x - 2\sin x$ on $[0, \pi]$

(c) $h(x) = \dfrac{x}{x^2 + 9}$ on $[0, 9]$

(d) The function of 6(b), $g(x) = \left(x^2 - 4\right)^{2/3}$, on $[-1, 3]$.

Solution. The method, for a continuous f on a closed interval $[a, b]$: (1) the values of f at the critical points *inside* (a, b); (2) the values at the endpoints a and b; (3) the largest of these is the absolute maximum, the smallest the absolute minimum. It works because the Extreme Value Theorem promises both values, and Fermat's Theorem says that an extreme value inside the interval must sit at a critical point.

(a) f is a polynomial, continuous on $[0, 3]$. $f'(x) = 6x^2 - 30x + 24 = 6(x^2 - 5x + 4) = 6(x - 1)(x - 4)$: critical points 1 and 4. Only 1 lies in $(0, 3)$; the critical point 4 is outside the interval and plays no part. *Step 1:* $f(1) = 2 - 15 + 24 + 3 = 14$. *Step 2:* $f(0) = 3$, $f(3) = 54 - 135 + 72 + 3 = -6$. *Step 3:* the absolute maximum is 14, at $x = 1$; the absolute minimum is -6, at the endpoint $x = 3$. (A student who also writes $f(4) = -13$ and calls it the minimum has used a point that is not in the interval.)

(b) g is continuous. $g'(x) = 1 - 2\cos x$, which is 0 when $\cos x = \frac{1}{2}$; in $(0, \pi)$ that is $x = \frac{\pi}{3}$ only. *Step 1:* $g\left(\frac{\pi}{3}\right) = \frac{\pi}{3} - 2\sin\frac{\pi}{3} = \frac{\pi}{3} - \sqrt{3} \approx 1.047 - 1.732 = -0.685$. *Step 2:* $g(0) = 0$, $g(\pi) = \pi - 0 = \pi$. *Step 3:* the absolute maximum is π, at $x = \pi$; the absolute minimum is $\frac{\pi}{3} - \sqrt{3}$, at $x = \frac{\pi}{3}$.

(c) h is continuous on $[0, 9]$ (the denominator is never 0). By the quotient rule,

$$h'(x) = \frac{1 \cdot (x^2 + 9) - x \cdot 2x}{(x^2 + 9)^2} = \frac{9 - x^2}{(x^2 + 9)^2},$$

which is zero at $x = \pm 3$; only 3 is in $(0, 9)$. *Step 1:* $h(3) = \dfrac{3}{18} = \dfrac{1}{6}$. *Step 2:* $h(0) = 0$, $h(9) = \dfrac{9}{90} = \dfrac{1}{10}$. *Step 3:* the absolute maximum is $\frac{1}{6}$, at $x = 3$; the absolute minimum is 0, at $x = 0$. (Since $\frac{1}{10} < \frac{1}{6}$: the graph rises to $\frac{1}{6}$ and then falls, but only to $\frac{1}{10}$ by $x = 9$.)

(d) g is continuous on $[-1, 3]$ (a cube root of a polynomial). From 6(b) the critical points are -2, 0 and 2; in $(-1, 3)$: 0 and 2. *Step 1:* $g(0) = \sqrt[3]{16} \approx 2.52$ and $g(2) = 0$. *Step 2:* $g(-1) = \sqrt[3]{9} \approx 2.08$ and $g(3) = \sqrt[3]{25} \approx 2.92$. *Step 3:* the absolute maximum is $\sqrt[3]{25}$, at the endpoint $x = 3$; the absolute minimum is 0, at $x = 2$ — a critical point where g' *does not exist*. Solving only $g'(x) = 0$ would have given the candidates 0, -1, 3 and the wrong "minimum" $\sqrt[3]{9}$.

Part II · The Mean Value Theorem and the shape of a graph (3.2–3.3) · 50 minutes

Warm-up II · True or false? (justify with one sentence or one counterexample)

1. If f is continuous on $[a, b]$ and $f(a) = f(b)$, then there is a number c in (a, b) with $f'(c) = 0$.
2. If $f'(x) > 0$ for every x in (a, b), then f is increasing on (a, b).
3. If $f'(x) = 0$ for every x in the domain of f, then f is constant.
4. If the graph of f is concave upward on (a, b), then $f''(x) > 0$ for every x in (a, b).
5. If f'' is continuous near c, $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c.
6. If f is continuous at 2, $f'(x) < 0$ for $x < 2$ and $f'(x) > 0$ for $x > 2$, then f has a local minimum at 2.

7. For $f(x) = x^2$ on $[1, 3]$, the number c of the Mean Value Theorem is $c = 2$.
8. If $f'(x) = g'(x)$ for every x, then $f(x) = g(x)$ for every x.
9. If f is concave upward on an interval, its graph lies above its tangent lines there.
10. For $f(x) = x^3 - 9x$ on $[-3, 3]$ there is exactly one number c in $(-3, 3)$ with $f'(c) = 0$.

Solution.

1 False. Rolle's Theorem also needs f *differentiable* on (a, b). $f(x) = |x|$ on $[-1, 1]$ is continuous with $f(-1) = 1 = f(1)$, but $f'(x) = -1$ for $x < 0$, $f'(x) = 1$ for $x > 0$, and $f'(0)$ does not exist: no c with $f'(c) = 0$.

2 True. This is the Increasing/Decreasing Test, proved from the Mean Value Theorem: for $x_1 < x_2$ in (a, b), $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) > 0$.

3 False. The theorem says: $f' = 0$ on an *interval* makes f constant there. The domain of $f(x) = 2$ (for $x < 1$), $f(x) = 5$ (for $x > 1$) is not an interval; $f'(x) = 0$ at every point of it, and f is not constant. (It is constant on each of the two pieces.)

4 False. The Concavity Test goes from $f'' > 0$ to concave upward, not back. $f(x) = x^4$ lies above every tangent line, so it is concave upward on $\mathbb{R}$; but $f''(x) = 12x^2$ is 0 at $x = 0$.

5 True. This is the Second Derivative Test, part (b): near c the graph is concave downward and lies below its horizontal tangent at c.

6 True. This is the First Derivative Test, part (b), and its proof is the Mean Value Theorem: for $x < 2$ near 2, $f(2) - f(x) = f'(d)(2 - x) < 0$ with d in $(x, 2)$, and for $x > 2$ near 2, $f(x) - f(2) = f'(d)(x - 2) > 0$; so $f(x) > f(2)$ on both sides — a local minimum. (Continuity at 2 is needed for the theorem's first hypothesis on $[x, 2]$ and $[2, x]$: a jump at 2 could spoil it.)

7 True. The secant slope is $\dfrac{f(3) - f(1)}{3 - 1} = \dfrac{9 - 1}{2} = 4$, and $f'(c) = 2c = 4$ gives $c = 2$, which is in $(1, 3)$. Here c happens to be the midpoint, and for a parabola it always is (Challenge 2); for other functions it is not — for x^3 on $[-1, 2]$ the lecture found $c = 1$, not $\frac{1}{2}$.

8 False. Equal derivatives on an interval make $f - g$ *constant*, not zero: $f(x) = x^2$ and $g(x) = x^2 + 7$ have the same derivative $2x$ everywhere.

9 True. That is the definition of concave upward (and the Concavity Test tells us when it happens: $f'' > 0$).

10 False. $f(-3) = -27 + 27 = 0 = f(3)$, so Rolle's Theorem applies and promises at least one c; in fact $f'(x) = 3x^2 - 9 = 0$ gives $c = \pm\sqrt{3} \approx \pm 1.732$: *two* numbers, both in $(-3, 3)$. The theorem promises at least one c, never exactly one.

Problem 8 *Rolle's Theorem and the Mean Value Theorem: find c.*

In (a)–(c), check that the hypotheses hold, then find *every* number c that the theorem promises.

(a) Rolle's Theorem for $f(x) = \sin 2x$ on $[0, \pi]$.

(b) The Mean Value Theorem for $f(x) = \sqrt{x}$ on $[0, 25]$. (Is f differentiable at 0? Does it matter?)

(c) The Mean Value Theorem for $f(x) = \ln x$ on $[1, e]$.

(d) Let $f(x) = |x - 1|$ on $[0, 4]$. Show that there is *no* number c with $f(4) - f(0) = f'(c)(4 - 0)$. Why does this not contradict the Mean Value Theorem?

Solution.

(a) *Hypotheses:* $\sin 2x$ is continuous on $[0, \pi]$ and differentiable on $(0, \pi)$ (everywhere, in fact), and $f(0) = \sin 0 = 0$, $f(\pi) = \sin 2\pi = 0$: $f(0) = f(\pi)$. *The c:* $f'(x) = 2\cos 2x$, and $\cos 2x = 0$ exactly when $2x = \dfrac{\pi}{2} + n\pi$, i.e. $x = \dfrac{\pi}{4} + \dfrac{n\pi}{2}$. The numbers of this form in the open interval $(0, \pi)$ are $c = \dfrac{\pi}{4}$ and $c = \dfrac{3\pi}{4}$ (the next ones, $-\frac{\pi}{4}$ and $\frac{5\pi}{4}$, are outside). Two numbers — the theorem promises at least one.

(b) *Hypotheses:* $\sqrt{x}$ is continuous on $[0,25]$ (the square root is continuous on $[0,\infty)$), and differentiable on $(0,25)$: $f'(x)=\dfrac{1}{2\sqrt{x}}$ exists for every $x>0$. At $x=0$ the derivative does not exist (the tangent is vertical) — but the theorem asks for differentiability on the *open* interval $(0,25)$ only, so $\sqrt{x}$ qualifies. *The c:* the secant slope is $\dfrac{f(25)-f(0)}{25-0}=\dfrac{5-0}{25}=\dfrac{1}{5}$, and

$$f'(c)=\frac{1}{2\sqrt{c}}=\frac{1}{5}\iff\sqrt{c}=\frac{5}{2}\iff c=\frac{25}{4}.$$

$\frac{25}{4}=6.25$ is in $(0,25)$. The tangent at $\left(\frac{25}{4},\frac{5}{2}\right)$ is parallel to the chord from $(0,0)$ to $(25,5)$.

(c) *Hypotheses:* $\ln x$ is continuous on $[1,e]$ and differentiable on $(1,e)$, with $f'(x)=\dfrac{1}{x}$. *The c:* the secant slope is $\dfrac{\ln e-\ln 1}{e-1}=\dfrac{1-0}{e-1}=\dfrac{1}{e-1}$, and $\dfrac{1}{c}=\dfrac{1}{e-1}$ gives $c=e-1\approx 1.718$, which is in $(1,e)\approx(1,2.718)$.

(d) Proof. $f(0)=|-1|=1$ and $f(4)=|3|=3$, so $\dfrac{f(4)-f(0)}{4-0}=\dfrac{2}{4}=\dfrac{1}{2}$. On the other hand $f(x)=1-x$ for $x<1$ and $f(x)=x-1$ for $x>1$, so $f'(x)=-1$ for $x<1$ and $f'(x)=1$ for $x>1$, and at $x=1$ the derivative does not exist (slopes -1 and 1 from the two sides). The only values f' ever takes are -1 and 1, never $\frac{1}{2}$: there is no such c. □ This does not contradict the Mean Value Theorem, because its second hypothesis fails: f is not differentiable on the open interval $(0,4)$ — there is a corner at $x=1$. The theorem promises a c only when both hypotheses hold.

Problem 9 *What the Mean Value Theorem can prove.*

(a) Show that the equation $x^3+9x-3=0$ has *exactly one* real solution, and find an interval of length $\frac{1}{2}$ that contains it. (At least one: the Intermediate Value Theorem. At most one: Rolle's Theorem.)

(b) Show that, whatever the constant k, the equation $x^4+6x+k=0$ has *at most two* real solutions.

(c) A function f is differentiable on $\mathbb{R}$, $f(2)=-1$, and $f'(x)\le 3$ for every x. How large can $f(6)$ possibly be? Give a function for which that value is reached.

(d) Prove the identity $\cos^{-1}x+\cos^{-1}(-x)=\pi$ for $-1\le x\le 1$, by differentiating the left side.

Solution.

(a) Proof. Let $f(x)=x^3+9x-3$, a polynomial: continuous and differentiable on $\mathbb{R}$. *At least one solution.* $f(0)=-3<0$ and $f\left(\frac{1}{2}\right)=\frac{1}{8}+\frac{9}{2}-3=\frac{13}{8}>0$. Since f is continuous on $\left[0,\frac{1}{2}\right]$ and 0 lies between $f(0)$ and $f\left(\frac{1}{2}\right)$, the Intermediate Value Theorem gives a number r in $\left(0,\frac{1}{2}\right)$ with $f(r)=0$. So $\left(0,\frac{1}{2}\right)$ is an interval of length $\frac{1}{2}$ that contains a solution. *At most one solution.* Suppose there were two, $a<b$, so $f(a)=0=f(b)$. Then f satisfies the three hypotheses of Rolle's Theorem on $[a,b]$, and there would be a number c in (a,b) with $f'(c)=0$. But $f'(x)=3x^2+9\ge 9>0$ for every x: f' is never 0. This contradiction shows that there cannot be two solutions. Together: exactly one real solution. □ (Numerically it is $r\approx 0.3294$.)

(b) Proof. Let $f(x)=x^4+6x+k$ and suppose it had three real zeros $r_1<r_2<r_3$. Rolle's Theorem on $[r_1,r_2]$ gives c_1 in (r_1,r_2) with $f'(c_1)=0$, and on $[r_2,r_3]$ a c_2 in (r_2,r_3) with $f'(c_2)=0$; and $c_1<r_2<c_2$, so $c_1\ne c_2$. Thus f' would have two zeros. But $f'(x)=4x^3+6$ (the constant k has disappeared), and $4x^3+6=0$ means $x^3=-\frac{3}{2}$, which has exactly one real solution, $x=-\sqrt[3]{3/2}\approx -1.145$, because x^3 is increasing and so never takes a value twice. A contradiction: there are at most two real zeros, whatever k is. □ (For $k=-7$ there are exactly two: $x=1$ is one, since $1+6-7=0$, and there is another near -2.1; for $k=100$ there are none. "At most two" covers both.)

(c) f is differentiable on $\mathbb{R}$, so the Mean Value Theorem applies on $[2,6]$: there is a c in $(2,6)$ with

$$f(6)-f(2)=f'(c)\,(6-2)=4f'(c)\le 4\cdot 3=12.$$

Hence $f(6) \le f(2) + 12 = -1 + 12 = 11$. The value 11 is reached: $f(x) = 3x - 7$ has $f(2) = -1$, $f'(x) = 3 \le 3$ everywhere, and $f(6) = 18 - 7 = 11$. So the answer is: $f(6)$ can be at most 11, and 11 is possible.

(d) Proof. Let $F(x) = \cos^{-1} x + \cos^{-1}(-x)$ on $[-1, 1]$. For $-1 < x < 1$, by the chain rule on the second term (inside $-x$, derivative -1),

$$F'(x) = -\frac{1}{\sqrt{1-x^2}} + \left(-\frac{1}{\sqrt{1-(-x)^2}}\right) \cdot (-1) = -\frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}} = 0.$$

A function whose derivative is 0 on the interval $(-1, 1)$ is constant there (the theorem proved from the Mean Value Theorem), so $F(x) = F(0) = \cos^{-1} 0 + \cos^{-1} 0 = \frac{\pi}{2} + \frac{\pi}{2} = \pi$ for $-1 < x < 1$. At the endpoints, directly: $F(1) = \cos^{-1} 1 + \cos^{-1}(-1) = 0 + \pi$ and $F(-1) = \pi + 0$, both π. $\square$ (The identity can also be seen without calculus: if $\theta = \cos^{-1} x$, then $\cos(\pi - \theta) = -\cos\theta = -x$ and $\pi - \theta \in [0, \pi]$, so $\cos^{-1}(-x) = \pi - \theta$. The calculus proof is the one that generalizes: whenever two functions have the same derivative on an interval, they differ by a constant, and one value fixes the constant.)

Problem 10 *Increasing or decreasing; the First Derivative Test.*

For each function find the intervals on which it is increasing or decreasing, and its local maximum and minimum values.

(a) $f(x) = 2x^3 - 15x^2 + 24x + 3$ (the function of 7(a), now on the whole line)

(b) $g(x) = x - \sqrt{2}\sin x$, $0 \le x \le 2\pi$

(c) $h(x) = x^4 - 6x^2 + 8x$ (factor h' completely; one factor is a square)

(d) $k(x) = x^2 - 8\ln x$

Solution. The method: find f', factor it, find where it is 0 or undefined, and make a sign chart; the I/D Test turns each sign into "increasing" or "decreasing", and the First Derivative Test turns each sign *change* at a critical point into a local maximum ($+ \to -$) or minimum ($- \to +$) — and no change into "neither".

(a) $f'(x) = 6x^2 - 30x + 24 = 6(x-1)(x-4)$. Sign chart:

interval	$x-1$	$x-4$	$f'(x)$	f
$x < 1$	$-$	$-$	$+$	increasing on $(-\infty, 1)$
$1 < x < 4$	$+$	$-$	$-$	decreasing on $(1, 4)$
$x > 4$	$+$	$+$	$+$	increasing on $(4, \infty)$

At 1, f' changes from $+$ to $-$: a local maximum, $f(1) = 14$. At 4, from $-$ to $+$: a local minimum, $f(4) = 128 - 240 + 96 + 3 = -13$. (On $[0, 3]$, in 7(a), the absolute minimum was at the endpoint 3, not at 4: on the whole line the picture is different.)

(b) $g'(x) = 1 - \sqrt{2}\cos x$, which is 0 when $\cos x = \frac{1}{\sqrt{2}}$: in $[0, 2\pi]$, $x = \frac{\pi}{4}$ and $x = \frac{7\pi}{4}$. On $\left(0, \frac{\pi}{4}\right)$ and on $\left(\frac{7\pi}{4}, 2\pi\right)$ the cosine is larger than $\frac{1}{\sqrt{2}}$, so $g' < 0$; on $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$ it is smaller (it even goes negative), so $g' > 0$. Hence g is decreasing on $\left(0, \frac{\pi}{4}\right)$ and on $\left(\frac{7\pi}{4}, 2\pi\right)$, increasing on $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$. Local minimum: $g\left(\frac{\pi}{4}\right) = \frac{\pi}{4} - \sqrt{2}\sin\frac{\pi}{4} = \frac{\pi}{4} - \sqrt{2}\cdot\frac{\sqrt{2}}{2} = \frac{\pi}{4} - 1 \approx -0.215$. Local maximum: $g\left(\frac{7\pi}{4}\right) = \frac{7\pi}{4} - \sqrt{2}\sin\frac{7\pi}{4} = \frac{7\pi}{4} + 1 \approx 6.498$. (The endpoint values $g(0) = 0$ and $g(2\pi) = 2\pi$ are not local extrema.)

(c) $h'(x) = 4x^3 - 12x + 8 = 4(x^3 - 3x + 2)$. Since $x = 1$ is a root of $x^3 - 3x + 2$, divide: $x^3 - 3x + 2 = (x-1)(x^2 + x - 2) = (x-1)(x-1)(x+2)$, so

$$h'(x) = 4(x-1)^2(x+2).$$

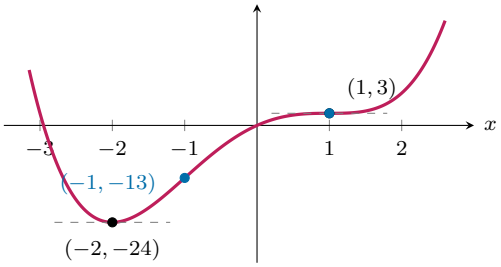
Critical points: -2 and 1. The factor $(x-1)^2$ is ≥ 0 everywhere and 0 only at 1, so the sign of h' is the sign of $x + 2$, except that $h'(1) = 0$:

interval	$(x-1)^2$	$x+2$	$h'(x)$	h
$x<-2$	$+$	$-$	$-$	decreasing on $(-\infty,-2)$
$-2<x<1$	$+$	$+$	$+$	increasing on $(-2,1)$
$x>1$	$+$	$+$	$+$	increasing on $(1,\infty)$

At -2, h' changes from $-$ to $+$: a local minimum, $h(-2)=16-24-16=-24$. At 1, h' is positive on both sides: *no* local maximum or minimum, although $h'(1)=0$ — the graph has a horizontal tangent at $(1,3)$ and keeps rising through it. (Since h is increasing on $(-2,1]$ and on $[1,\infty)$, it is increasing on all of $(-2,\infty)$.)

(d) The domain is $x>0$ (because of $\ln x$). $k'(x)=2x-\dfrac{8}{x}=\dfrac{2x^2-8}{x}=\dfrac{2(x-2)(x+2)}{x}$. In the domain the only zero is $x=2$ ($x=-2$ is not in the domain and is *not* a critical point). For $0<x<2$: $x-2<0$, $x+2>0$, $x>0$, so $k'<0$; for $x>2$: $k'>0$. So k is decreasing on $(0,2)$ and increasing on $(2,\infty)$, with a local minimum $k(2)=4-8\ln 2\approx -1.545$ and no local maximum. (The local minimum is also the absolute minimum on the domain.)

10(c), 11(a): $y=x^4-6x^2+8x$



Problem 11 *Concavity, inflection points and the Second Derivative Test.*

(a) For $h(x)=x^4-6x^2+8x$ (from 10(c)): find the intervals of concavity and the inflection points; then apply the Second Derivative Test at each critical point — and say what to do where it gives no information.

(b) $g(x)=x\,e^{6x}$: concavity, inflection points, and the local extreme values by the Second Derivative Test.

(c) Let $h(x)=x\tan^{-1}x$. Show that $h''(x)=\dfrac{2}{(1+x^2)^2}$, and deduce the concavity of the graph and the minimum value of h.

Solution.

(a) From 10(c), $h'(x)=4x^3-12x+8$, so $h''(x)=12x^2-12=12(x-1)(x+1)$: positive for $x<-1$, negative for $-1<x<1$, positive for $x>1$. By the Concavity Test the graph is concave upward on $(-\infty,-1)$ and on $(1,\infty)$ and concave downward on $(-1,1)$. The concavity changes at $x=-1$ and at $x=1$, where h is continuous: the inflection points are $\big(-1,h(-1)\big)=(-1,1-6-8)=(-1,-13)$ and $\big(1,h(1)\big)=(1,1-6+8)=(1,3)$. *The Second Derivative Test* at the critical points -2 and 1 (where $h'=0$): $h''(-2)=12\cdot(-3)\cdot(-1)=36>0$, so h has a local minimum at -2, $h(-2)=-24$ — the same verdict as the First Derivative Test. $h''(1)=0$: the test gives *no information*. What to do: go back to the First Derivative Test — in 10(c), h' is positive on both sides of 1, so there is neither a maximum nor a minimum there. In fact $(1,3)$ is an inflection point with a horizontal tangent: the curve crosses its tangent line there. (The picture under Problem 10.)

(b) Product rule: $g'(x)=1\cdot e^{6x}+x\cdot 6e^{6x}=(1+6x)e^{6x}$, and again $g''(x)=6e^{6x}+(1+6x)\cdot 6e^{6x}=(6+6+36x)e^{6x}=12(3x+1)e^{6x}$. Since $e^{6x}>0$, the sign of g'' is the sign of $3x+1$: negative for $x<-\frac{1}{3}$, positive for $x>-\frac{1}{3}$. Concave downward on $\big(-\infty,-\frac{1}{3}\big)$, upward on $\big(-\frac{1}{3},\infty\big)$; the inflection point is $\big(-\frac{1}{3},g(-\frac{1}{3})\big)=\big(-\frac{1}{3},-\frac{1}{3}e^{-2}\big)=\big(-\frac{1}{3},-\frac{1}{3e^2}\big)\approx(-0.333,-0.045)$. *Local extreme values:* $g'(x)=0$ exactly when $1+6x=0$, i.e. $x=-\frac{1}{6}$ (the only critical point). $g''\big(-\frac{1}{6}\big)=12\cdot\big(-\frac{1}{2}+1\big)\cdot e^{-1}=\dfrac{6}{e}>0$:

by the Second Derivative Test, a local minimum, $g\left(-\frac{1}{6}\right) = -\frac{1}{6}e^{-1} = -\frac{1}{6e} \approx -0.061$. There is no local maximum.

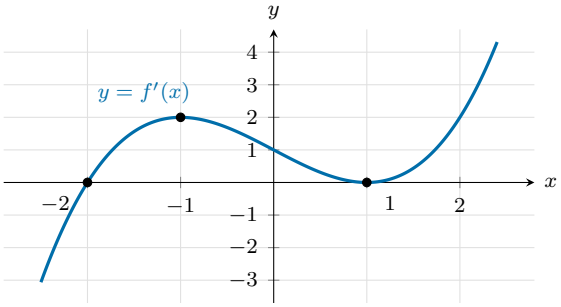
(c) Proof. Product rule: $h'(x) = 1 \cdot \tan^{-1} x + x \cdot \dfrac{1}{1+x^2} = \tan^{-1} x + \dfrac{x}{1+x^2}$. Then, with the quotient rule on the second term,

$$h''(x) = \frac{1}{1+x^2} + \frac{1 \cdot (1+x^2) - x \cdot 2x}{(1+x^2)^2} = \frac{1}{1+x^2} + \frac{1-x^2}{(1+x^2)^2} = \frac{(1+x^2)+(1-x^2)}{(1+x^2)^2},$$

and the numerator is 2: $h''(x) = \dfrac{2}{(1+x^2)^2}$. □ This is positive for every x: the graph is concave upward on the whole line, and there are no inflection points. *The minimum:* $h'(0) = \tan^{-1} 0 + 0 = 0$, so 0 is a critical point, and $h''(0) = 2 > 0$: a local minimum by the Second Derivative Test, with value $h(0) = 0$. It is the absolute minimum: $h'' > 0$ everywhere makes h' increasing, so $h'(x) < h'(0) = 0$ for $x < 0$ and $h'(x) > 0$ for $x > 0$ — h decreases up to 0 and increases after it. (Directly: x and $\tan^{-1} x$ have the same sign, so $x \tan^{-1} x \geq 0$, with equality only at 0.)

Problem 12 *Reading f'; sketching from conditions.*

(a) The graph of the *derivative* f' of a function f is shown, for $-2.5 \leq x \leq 2.4$ (not the graph of f). On what intervals is f increasing, and on what intervals decreasing? At which x does f have a local maximum or minimum? On what intervals is the graph of f concave upward or downward, and at which x are its inflection points?



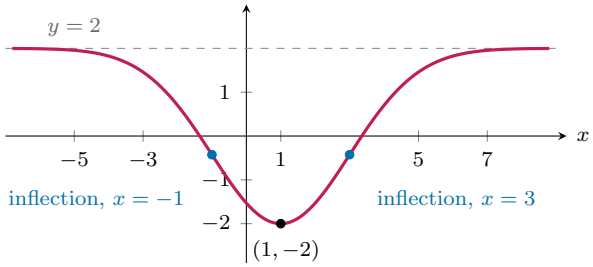
(b) Sketch a possible graph of a function f that satisfies all of the following: (i) $f(1) = -2$ and $f'(1) = 0$; (ii) $f'(x) < 0$ for $x < 1$ and $f'(x) > 0$ for $x > 1$; (iii) $f''(x) > 0$ for $-1 < x < 3$, and $f''(x) < 0$ for $x < -1$ and for $x > 3$; (iv) $\lim\limits_{x \to -\infty} f(x) = 2 = \lim\limits_{x \to \infty} f(x)$. What kind of point is $(1, -2)$, and what happens at $x = -1$ and $x = 3$?

Solution.

(a) Everything about f must be read from the *sign* and the *slope* of the curve shown, which is f'. *Where f' is positive or negative:* the curve is below the x-axis for $-2.5 < x < -2$, crosses it at $x = -2$, is above it for $-2 < x < 1$, touches it at $x = 1$ without crossing, and is above it again for $1 < x < 2.4$. By the I/D Test, f is decreasing on $(-2.5, -2)$ and increasing on $(-2, 1)$ and on $(1, 2.4)$ — in fact on all of $(-2, 2.4)$. *Local extrema:* f' changes from negative to positive at $x = -2$: a local minimum of f there (First Derivative Test). At $x = 1$, $f'(1) = 0$, but f' is positive on both sides: no local maximum or minimum — the graph of f has a horizontal tangent at $x = 1$ and keeps rising. (The values $f(-2)$ and $f(1)$ cannot be read from this picture; only the x's.) *Concavity:* f'' is the derivative of f' — the *slope* of the curve shown. That slope is positive for $-2.5 < x < -1$ (the curve rises), zero at $x = -1$ (its high point), negative for $-1 < x < 1$ (it falls to 0 at $x = 1$), zero at $x = 1$ (a horizontal tangent of f'), and positive for $1 < x < 2.4$. So $f'' > 0$ on $(-2.5, -1)$ and on $(1, 2.4)$, $f'' < 0$ on $(-1, 1)$: the graph of f is concave upward on $(-2.5, -1)$ and on $(1, 2.4)$, concave downward on $(-1, 1)$, and the inflection points are at $x = -1$ and $x = 1$ — where f'' changes sign, i.e. where the curve shown has its local maximum and its local minimum. (The curve shown is $f'(x) = \frac{1}{2}(x+2)(x-1)^2$; one function with this derivative is $f(x) = \frac{x^4}{8} - \frac{3x^2}{4} + x$, and any other differs from it by a constant.)

(b) Build the sketch condition by condition. (ii): f decreases on $(-\infty, 1)$ and increases on $(1, \infty)$, so (i) places a horizontal tangent at the lowest point $(1, -2)$: a local minimum — in fact the absolute minimum — by the First Derivative Test. (The Second Derivative Test would give the same verdict from $f''(1) > 0$, but it also needs f'' continuous near 1, which the conditions do not say.) (iv): the graph has the horizontal asymptote $y = 2$ on both sides; since f decreases towards the minimum from the left and increases away from it on the right, the curve approaches $y = 2$ from *below* at both ends. (iii): concave upward on $(-1, 3)$ — the bowl around the minimum — and concave downward outside, which is exactly what bending over towards an asymptote from below looks like; the concavity changes at $x = -1$ and at $x = 3$, so those are the inflection points. One function with all these properties is $f(x) = 2 - 4e^{-(x-1)^2/8}$ (the picture); yours need not be this one.

12(b): one possible graph, $y = 2 - 4e^{-(x-1)^2/8}$



Challenge *(for those who finish early, or at home)*

1. Use the Mean Value Theorem to show that $\left|\tan^{-1} a - \tan^{-1} b\right| \le |a - b|$ for all real numbers a and b.
2. Let $f(x) = px^2 + qx + r$ with $p \neq 0$. Show that, on *any* interval $[a, b]$, the number c of the Mean Value Theorem is the midpoint $\dfrac{a+b}{2}$ — and give a function for which it is not.
3. The inverse cotangent is defined by $\cot^{-1} x = y \iff \cot y = x$ and $0 < y < \pi$ (for every real x). Prove that $\dfrac{d}{dx}\cot^{-1} x = -\dfrac{1}{1+x^2}$, as the lecture proved the formula for $\tan^{-1}$.

Solution.

1. Proof. If $a = b$ both sides are 0. If $a \neq b$, say $b < a$, then $\tan^{-1}$ is continuous on $[b, a]$ and differentiable on (b, a), so the Mean Value Theorem gives a c in (b, a) with

$$\tan^{-1} a - \tan^{-1} b = \left(\tan^{-1}\right)'(c)\,(a - b) = \frac{1}{1+c^2}\,(a - b).$$

Since $1 + c^2 \ge 1$, $0 < \dfrac{1}{1+c^2} \le 1$, and taking absolute values, $\left|\tan^{-1} a - \tan^{-1} b\right| \le |a - b|$. (For $a < b$ swap the names.) □

2. Proof. $\dfrac{f(b) - f(a)}{b - a} = \dfrac{p(b^2 - a^2) + q(b - a)}{b - a} = p(a+b) + q$, and $f'(c) = 2pc + q$. So $f'(c)$ equals the secant slope exactly when $2pc = p(a+b)$, i.e. (as $p \neq 0$) $c = \dfrac{a+b}{2}$: for a parabola the tangent parallel to a chord is always at the midpoint of the chord. □ For other functions it is not: for $f(x) = x^3$ on $[-1, 2]$ the number is $c = 1$ (the secant slope is 3 and $3c^2 = 3$ with c in $(-1, 2)$), while the midpoint is $\frac{1}{2}$.

3. Proof. On $(0, \pi)$, cot is one-to-one (decreasing) and differentiable, with $\cot' y = -\csc^2 y \neq 0$ there; so by the theorem on the derivative of an inverse function, $\cot^{-1}$ is differentiable on $\mathbb{R}$. Let $y = \cot^{-1} x$, so $\cot y = x$ with $0 < y < \pi$. Differentiate implicitly: $-\csc^2 y \cdot y' = 1$, so

$$y' = -\frac{1}{\csc^2 y} = -\frac{1}{1 + \cot^2 y} = -\frac{1}{1+x^2},$$

using $1 + \cot^2 y = \csc^2 y$. □ (Consistent with $\cot^{-1} x = \dfrac{\pi}{2} - \tan^{-1} x$, whose derivative is $-\dfrac{1}{1+x^2}$.)